InvestCircle: A Peer-to-Peer Stock Research and Community Platform

Project Proposal for UCS503P

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1 Higher-order Goal

The higher-order goal of InvestCircle is to create a more accountable and transparent environment for peer-to-peer financial research. Retail investors currently depend on social media, discussion forums, financial news websites and informal stock-tip communities, where opinions are often fragmented and historical predictions are rarely tracked.

InvestCircle aims to translate the idea of *accountable financial discussion* into a measurable software system. Instead of evaluating a contributor only through followers, likes or engagement, the platform maintains a persistent history of structured stock calls and their eventual outcomes.

The project is focused on financial research and discussion rather than trading. It will not execute trades or connect users to brokerage accounts.

2 Time-to-Value

The project will prioritize the core user workflow and deliver a functional system incrementally.

- The first working version will focus on authentication, user profiles, stock discussions, watchlists and structured stock calls.
- The call-tracking mechanism will be implemented early because it is the main feature differentiating InvestCircle from ordinary discussion platforms.
- Development will follow short iterations so that individual modules can be tested independently before integration.
- External market-data integration will be introduced after the core database and API workflow is stable.
- Optional features such as AI tagging, AI-inferred call detection and video reactions will only be implemented after the core platform is functional.

The first meaningful milestone is therefore a working platform where a user can register, publish a structured stock call, follow another user, maintain a watchlist and eventually view the resolved outcome of the call.

3 Problem Statement

Retail investors commonly combine financial news websites, social-media discussions and informal stock-tip communities when researching investments. Although these platforms provide large amounts of information, they have several limitations.

- **Lack of accountability:** users can repeatedly publish stock predictions without maintaining a visible record of previous outcomes.
- **Fragmented research:** financial news, stock discussions and individual research are distributed across multiple platforms.
- **Unstructured opinions:** many investment opinions are written as free-form text, making explicit predictions difficult to identify and evaluate.
- **Weak credibility signals:** follower counts and engagement do not necessarily represent historical prediction accuracy.
- **Limited personalization:** users interested in specific stocks or sectors must often manually search for relevant news.
- **Poor historical visibility:** unsuccessful predictions may disappear from attention while successful predictions remain visible.
- **Doomscrolling without substance:** short-form scrolling formats (reels) are highly engaging but rarely used for anything meaningful, while genuinely useful discussion platforms are rarely built to be scrollable.

InvestCircle addresses these problems through identity-based participation, structured call tracking, public performance history, watchlists and personalized, scrollable financial content in one platform.

4 Proposed System

4.1 Overview

InvestCircle is a web-based peer-to-peer financial research platform combining community discussion with market information. It is organized around three primary sections: a **Discussions** tab for Reddit-style, identity-based stock discussion; a **News Reel** for a fast, scrollable, reels-style feed of live market news; and a **Personal** page where each user's own posts, participation and track record live.

The main components are:

- **User and identity system:** registration, authentication, profiles, followers and following.
- **Discussion platform:** research posts, voting, comments and threaded replies, under a real identity rather than an anonymous handle.
- **Structured stock calls:** ticker, direction, target price and time horizon, attached to a discussion post.
- **Call tracking:** automatic evaluation of completed calls using market-price data.
- **Watchlists / portfolio:** tickers a user manually adds because they are interested in them; used both for personal organization and to personalize the News Reel.
- **News Reel:** a continuously scrollable, reels-style market news feed personalized using watchlist interests.

- **Notifications:** updates for follows, comments and resolved calls.
- **Search and discovery:** search across users, stocks and discussions.
- **Optional AI features:** sector/sentiment tagging of posts, and AI-inferred detection of implied stock calls from free-text discussion.

4.2 Core Workflow

The main workflow is designed around the creation and evaluation of a stock call.

1. A registered user creates a discussion post.
2. The user may attach a structured stock call containing a ticker, bullish/bearish direction, target price and time horizon.
3. The system stores the call with a *Pending* status.
4. When the specified horizon expires, a background process obtains the relevant market price.
5. The system checks whether the target price was reached.
6. The call is marked as *Hit* or *Miss*.
7. The result becomes part of the user's public historical track record, shown on their Personal page.

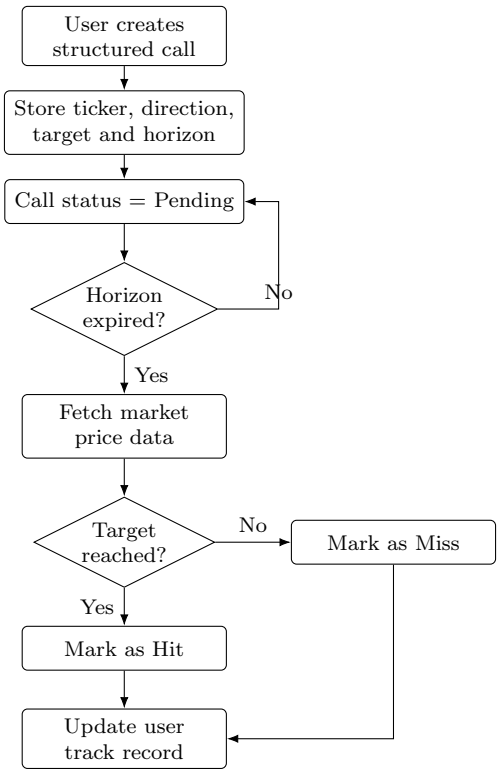


Figure 1: Stock-call resolution workflow used to generate the contributor track record.

4.3 User and Identity System

Users register using email/password or OAuth. Passwords are never stored directly and are protected using a modern password-hashing mechanism such as bcrypt or Argon2.

Each user has a persistent profile (the Personal page) containing identity information, biography, followers, following count, their own posts and questions, and their historical stock-call results with an overall accuracy score. The identity-based model is central to the accountability mechanism because each structured call remains associated with its original author, and discussion is deliberately not anonymous.

4.4 Discussion System

The Discussions tab is a Reddit-style feed of research notes, questions and stock opinions, always tied to a real user identity rather than a handle. Posts can contain a stock ticker, research text, an optional structured call, and supporting discussion. Users can upvote, downvote, comment and reply to discussions. Comments use a self-referencing relationship so that replies can form discussion threads.

4.5 News Reel and Watchlists

The News Reel is a fast, vertically scrolling, reels-style feed of live market news, aimed at replacing unproductive scrolling with something useful. Each news item may contain a headline, source, publication timestamp, related ticker and market category, obtained from an external financial-data provider.

Users maintain watchlists containing stocks they are manually interested in. These watchlists serve both as personal organization and as a signal to personalize the News Reel, so that news related to a user's followed tickers and sectors is surfaced more prominently. The initial implementation will use simple, deterministic ranking (boosting items that match a user's watchlist) rather than a complex recommendation model.

As an optional extension, a user may record a short video reaction to a news item (*"Share Your Opinion"*). Other users viewing the same news item can see up to three video reactions from other users, prioritized by verified status and follower count.

4.6 Optional Social and AI Features

Users can follow contributors and receive notifications for important activities such as new followers, comments and call resolutions.

Two optional AI components are considered, kept outside the platform's critical path:

- **Sector/sentiment tagging:** classifies posts into broad sector and sentiment categories (positive, neutral, negative) at creation time.
- **AI-inferred call detection:** reads free-text discussion posts to detect an implied stock opinion (ticker plus bullish/bearish stance) even when the user has not filled in a structured call. Such inferred calls are checked against real price data after a set time window, the same way explicit calls are, but are labelled *AI-inferred* wherever shown, so they are never silently mixed into a user's confirmed track record.

Authentication, discussions, watchlists and explicit call tracking must remain fully functional even when both AI components are unavailable.

5 Solution Approach

This is primarily a software-engineering project. The focus is on system correctness, modularity, workflow automation, scalability and measurable evaluation rather than developing novel financial prediction algorithms.

5.1 Web Architecture

A layered architecture will separate the major system responsibilities.

- **Frontend:** React-based responsive web interface for discussions, news, watchlists and the personal page, using a clean, minimal visual style with a light and a dark mode.
- **Backend API:** REST endpoints for authentication, users, posts, comments, calls, watchlists and notifications.
- **Business Logic:** call resolution, feed ranking, follower relationships and notification generation.
- **Database:** PostgreSQL for persistent relational data.
- **External Services:** financial-market APIs and optional AI services.

5.2 Database Design

The principal entities are:

- `users`
- `posts`
- `comments`
- `follows`
- `watchlists`
- `watchlist_items`
- `calls`
- `notifications`
- `news_items`
- `reactions`

A post can optionally reference one structured call. Comments use `parent_comment_id` to represent threaded replies. Calls carry a flag distinguishing an explicit, user-entered call from an AI-inferred one.

Foreign keys maintain referential integrity while indexes are applied to frequently queried attributes such as user IDs, tickers and timestamps.

5.3 Background Processing and Caching

Background processing will handle operations that do not need to block user requests. These include:

- Market-data synchronization
- Expired-call resolution
- Notification generation
- News-feed updates

Redis can be used to cache market information and frequently requested feed data. Pagination or cursor-based retrieval will prevent the frontend from loading the complete history of large feeds.

5.4 Authentication and Security

Security will be implemented using secure password hashing, token-based authentication, authorization and API-level input validation. Users can modify only resources they own, while privileged operations require appropriate permissions. Database access will use parameterized queries or an ORM instead of constructing SQL queries directly from user input. Rate limiting can be applied to login, posting and reporting operations.

5.5 CI/CD and Testing

The development process will use automated testing and continuous integration.

Testing will include:

- Unit testing of call-resolution and feed-ranking logic.
- Integration testing of APIs and database operations.
- Frontend testing of critical user workflows.
- Load testing of feed and profile endpoints.

The target is at least 70% automated coverage for backend and business-logic components.

6 Evaluation Criteria

The system will be evaluated using measurable software and workflow metrics.

6.1 Primary Evaluation Metrics

- **Call-resolution reliability:** at least 95% of due calls should be automatically resolved without manual intervention.
- **Feed performance:** discussion and news feeds should load within approximately 1.5 seconds under seeded project-scale workload.
- **Test coverage:** backend and business-logic components should achieve at least 70% automated test coverage.

6.2 Secondary Evaluation Metrics

- All Must-priority functional requirements should be demonstrated end-to-end.
- Core functionality should remain available when optional AI services are unavailable.
- Database queries should maintain acceptable response time under increasing seeded data volume.
- The call history displayed for a contributor should correspond to stored call-resolution records.

7 Scalability

The architecture is designed to support growth in users, discussions, watchlists and market data.

1. **Stateless backend:** API servers will avoid storing application state locally wherever possible, allowing multiple instances to run concurrently.
2. **Database indexing:** indexes will be created for common searches and feed queries.
3. **Pagination:** large feeds will use pagination or cursor-based retrieval.

4. **Caching:** Redis can reduce repeated requests for market data and frequently accessed content.
5. **Asynchronous processing:** background workers will handle market synchronization, notifications and call resolution.
6. **Provider abstraction:** financial-data APIs will be isolated behind a service layer so that providers can be replaced if required.

Operationally, the application can be deployed using common cloud hosting, managed PostgreSQL and containerized backend services.

8 Engine Availability Heuristic

InvestCircle can be implemented using mature and widely available software components.

- React for the web frontend.
- Node.js and a REST-capable framework for backend services.
- PostgreSQL for relational persistence.
- Redis for caching and background-job support.
- JWT and secure password hashing for authentication.
- Alpha Vantage, Finnhub or another suitable financial-data provider.
- Standard unit and integration testing frameworks.
- Git-based version control and CI/CD automation.

These technologies allow the team to focus on workflow design, correctness and accountability rather than developing infrastructure from scratch.

9 Project Scope and Deliverables

9.1 Initial Deliverable

The first working version will provide:

- User registration and authentication.
- User profiles (Personal page).
- Stock discussion posts (Discussions tab).
- Structured stock calls.
- Follow and unfollow functionality.
- Basic watchlists.
- Call status and resolution storage.
- PostgreSQL database integration.
- REST API for the core workflow.

9.2 Subsequent Deliverables

Later iterations will add:

- Automatic market-price-based call resolution.
- Personalized News Reel driven by watchlists.
- Comments and threaded replies.
- Notifications.
- Search and discovery.
- Contributor performance statistics.
- Optional sector and sentiment tagging.
- Basic moderation functionality.

9.3 Optional Extensions

The following features are outside the core MVP and will be implemented only if sufficient development capacity remains:

- Short video reactions to financial news.
- AI-inferred call detection from free-text discussion.
- More advanced recommendation algorithms.
- Advanced analytics and contributor rankings.

10 Risks and Mitigations

- **Risk:** External API dependency — market-data providers may impose rate limits or experience downtime.
  Mitigation: Caching, scheduled synchronization and provider abstraction will reduce this dependency.
- **Risk:** Incorrect call resolution — inconsistent market data could affect results.
  Mitigation: The system will store resolution timestamps and the data source used for evaluation.
- **Risk:** Scope creep — AI features, video reactions and advanced recommendations can increase development effort.
  Mitigation: They will remain isolated from the core MVP.
- **Risk:** Security risks — unauthorized account access.
  Mitigation: Mitigated through password hashing, authentication, authorization, validation and rate limiting.
- **Risk:** Performance degradation — increasing data volume can slow feeds and searches.
  Mitigation: Pagination, indexing and caching will be introduced from the beginning.
- **Risk:** Uneven team contribution — development effort may not be evenly distributed across the team.
  Mitigation: Development will be divided into explicit module ownership and monthly milestones.

11 Summary

InvestCircle is a web platform for peer-to-peer stock research where discussion happens under real identities, not anonymous handles. Every structured stock call a user makes is automatically checked against real market data and rolled into a public, permanent accuracy record, so credibility is earned rather than claimed.

The platform is organized around three main sections: a Reddit-style discussion feed, a fast, scrollable feed of live market news, and a personal page where each user's own posts, participation and track record are visible to others. User-maintained watchlists double as an interest signal that personalizes the news feed toward the stocks and sectors a user actually cares about.

The project is being developed by a four-person team over four months, with the core discussion-and-accountability workflow built and demonstrated first, and additional features layered on only once that core is stable.